

Yuelin Liu

FULL-STACK DEVELOPER · AI & LLM APPLICATIONS

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SUMMARY

Early-career full-stack developer focused on AI/LLM applications and reliable data-driven workflows. Implemented OpenAI function-calling in an early-stage SaaS codebase; built MoneyGuard, a TypeScript vision-to-reasoning pipeline with Zod-validated model output and deterministic finance logic; and hardened a Next.js/MongoDB club system around server-side authorisation and regression-tested CI. Hands-on across TypeScript/Node.js, Python, databases and cloud-backed application delivery. Full Australian work rights until March 2029.

EXPERIENCE

Independent Developer

Sep 2025 – Present

Prospective Client Demos for a Melbourne Consultancy

Melbourne, Australia

- Build demonstration applications for prospective clients, including a specialty coffee retailer and a drone training provider, translating informal briefs into working prototypes for business pitches.
- Implement authenticated, database-backed prototypes with Next.js 15 App Router, Clerk OAuth and Convex as requirements become clearer.

Full-Stack Developer

Mar 2025 – Jun 2025

ByteCroniX – Early-Stage AI SaaS

Melbourne, Australia

- Implemented OpenAI function-calling flows in a Node.js/Express codebase for AI-assisted product interactions.
- Added timing instrumentation to an inherited points-summary path and identified current/proposed scoring services as the dominant latency contributors.
- Profiled an inherited points-summary pipeline and parallelised independent service calls, reducing an observed local integration-test response from 26.7s to 5.6s.

SELECTED PROJECTS

MoneyGuard | *TypeScript, Node.js, Gemini, DeepSeek, Zod* | case study | repo

- Built a timecard-to-wage pipeline that sends the image to Gemini for structured OCR, validates model output with Zod, computes wage and burn metrics deterministically, and streams a written audit from DeepSeek.
- Separated vision, reasoning and deterministic computation behind provider interfaces; deterministic mock providers run the end-to-end pipeline without API keys or network access.
- Added a protected totals-only extraction endpoint with bearer authentication, bounded upload checks and an explicit hosted-data boundary distinct from the local CLI/library workflow.

Melbourne University Ultimate Club Platform | *Next.js 15, TypeScript, MongoDB, Vitest* | repo

- Revisited and hardened a club-management application used for rosters, alumni records, events, announcements and video content, adding server-side authorisation and database-backed user/admin roles.
- Turned authorisation and write-path findings into automated regression tests; GitHub Actions runs typecheck, lint, tests and a production build on pushes and pull requests.

Client Data Delivery Validator | *Python, pandas, pytest* | repo

- Built a Python validation workflow for synthetic client-delivery CSV data with structured issue records and a small CLI; 12 pytest cases cover required fields, duplicate IDs, domain rules and invalid prices.

TECHNICAL SKILLS

Languages & Querying: TypeScript, JavaScript, Python, SQL (coursework)

AI & Validation: OpenAI function calling, Gemini, DeepSeek, Zod, structured model outputs, deterministic validation boundaries

Frontend: React 19, Next.js 15/16, Tailwind CSS, Recharts

Backend & Data: Node.js, Express, REST APIs, MongoDB, Convex, pandas, IndexedDB/Dexie

Authentication & Testing: Clerk OAuth, NextAuth, Vitest, Playwright, pytest, React Testing Library

Cloud & Delivery: GitHub Actions, Docker, AWS Lambda, Amazon SQS, Amazon Bedrock, Terraform

EDUCATION

The University of Melbourne

Master of Information Technology

2022 – 2023

- Selected coursework: Cryptography & Security (84, H1); Software Processes & Management (77, H2A); Database Systems & Information Modelling (73, H2B); Advanced Database Systems (68, H3).

University of Nottingham Ningbo China

BSc (Hons) Computer Science

2016 – 2020

APPLIED AI ARCHITECTURE STUDY

Agentic AI Engineering Course Implementation on AWS – Ed Donner (Udemy), completed Nov 2025

- Adapted the course's ALEX capstone locally to study an SQS-backed five-role portfolio-analysis workflow and Terraform-defined AWS architecture; deployed the study environment to AWS and later tore it down when the running cost outweighed its learning value.
- Worked with Amazon Bedrock, Lambda, SQS, Aurora Serverless v2 and Terraform, and contributed bounded fixes/documentation including a database-integrity verification step and a cross-region ECR deployment note.

ANALYTICAL & TECHNICAL WORK

Optimisation of Weighted Nearest Neighbours via Metric Learning and Batch Gradient Descent

- Sole-authored undergraduate dissertation using Python for metric-learning experiments, analytical MSE gradient derivation and repeatable evaluation across three UCI datasets. paper & code

Lattice-Based Post-Quantum Cryptography: Key Agreement Protocols on the LWE Problem

- Co-authored three-person COMP90043 Cryptography & Security course report (course result: 84, H1); authored the lattice preliminaries and LWE problem formulations. paper & code

ADDITIONAL INFORMATION

Work rights: Full Australian work rights – Temporary Graduate visa (subclass 485), valid until March 2029.

Role focus: AI/LLM application engineering, full-stack product engineering and backend integration roles where practical ownership, reliable data handling and continuous learning matter.